



Department of Information Technology

Academic Year 2025 – 2026(Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CC3391 & Object-Oriented Programming

Name of the Faculty member (s): Dr. K. Palraj, ASP / IT

Innovative Practice Description

- **Unit / Topic:** Unit 2 / Method Overloading
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO06.
- **Activity Chosen:** Flipped Class Room
- **Justification:**
 - In a flipped classroom, Method Overloading is highly effective because students can first learn the basic syntax and concept through youtube shared by the faculty, then use in-class time for hands-on activities that explore how the same method name can handle different parameter lists and data types. This approach promotes active learning, as students experiment with various overloads, analyze how Java resolves method calls, and apply the concept to real-world scenarios, thereby enhancing problem-solving skills, code flexibility, and higher-order thinking.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - In the flipped classroom approach for teaching Method Overloading, students are first learned about method overloading from the following youtube link given by the faculty <https://www.youtube.com/watch?v=km8xnlSRtWg>.
 - During the in-class session, the instructor guides students to progressively create overloaded methods by varying the number, type, and order of parameters for the same method name.
 - Students work individually or in groups to write, test, and debug overloaded methods, with the instructor facilitating discussions on method resolution, advantages, and practical applications.
 - The session concludes with peer review and a short reflective exercise to reinforce the learning outcomes.
 - Ms. Abinaya N explained the concept of method overloading in Figure 1.

• *CO – PO / PSO mapping:*

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	3	2	3	3	3	1	1	3	2	2

(1 – Low 2 – Moderate 3 – High)

• *PO / PSO mapped:*

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	3	2	3	3	3	1	1	3	2	2
Justification for correlation	Students will be able to apply OOP concepts like inheritance and interfaces to develop structured programs based on programming fundamentals	Students will be able to identify the suitable OOP features for a given problem involves problem decomposition and logical analysis.	Students will be able to design a solution for the problems using inheritance and packages encourage modular, reusable, and maintainable design	Students will be able to test and debug programs involving class hierarchies and interfaces to validate correctness and efficiency	Students will be able to practice using tools like code editors, compilers, and version control systems to support object-oriented program development.	Students practice teamwork by working together on object-oriented programming tasks and projects	Students will be able to explain their programs using comments, diagrams, and presentations to share their ideas clearly	Students learn to plan and complete small coding tasks like a project.	Students could be able to learning inheritance and interfaces builds a foundation for future learning in software development	Students will be able to uses OOP principles like inheritance and interfaces to solve structured problems efficiently	Students will be able to to inheritance and interfaces are foundational to development in technologies like Android, IoT, and Java-based enterprise applications	Students will be able to encourages clean architecture and modular code, aligning with best practices in software engineering and open-source contributions.

- **Images / Screenshot of the practice:**



Figure 1: Discussion on the topic by the team of Abinaya N

- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

- Students found the flipped classroom approach made Method Overloading easier to understand.
 - Students can report that real-world examples improved understanding of practical applications.
 - Students can enjoy peer discussions and group activities, which increased confidence and teamwork skills.

- ***Benefit of the practice:***

- Students can improve coding skills and accuracy through guided, hands-on practice.
 - Prepares students for industry expectations by simulating real coding challenges with multiple input types.
 - Students can increase their confidence and independence in applying Java concepts to practical problems.

- ❖ ***Challenges faced in implementation:***

- Some students with weak programming fundamentals required extra support to grasp overloading concepts.
 - Some students skipped pre-class preparation, leading to uneven participation levels during in-class activities.
 - Time management was challenging as group discussions sometimes extended beyond planned limits.

References:

- <https://www.ritrjpm.ac.in/departments/rit-information-technology/it-innovative-practices.php>
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/KP_CS3591_Flipped_Class_Room.pdf

Signature of Faculty Member

HOD